

KKS Suhith Babu

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Portfolio: kocherlasuhith12.github.io/portfolio-website

PROFESSIONAL SUMMARY

Final-year Computer Science student (Class of 2027) with hands-on experience in software engineering, backend development, AI engineering, automated testing, and cloud-native development. Skilled in Python, Java, FastAPI, Spring Boot, REST APIs, SQL/NoSQL databases, Docker, Git, CI/CD, and AWS. Experienced in building and deploying backend and full-stack systems with authentication, concurrency control, automated testing, and AI-powered features. Strong foundation in Object-Oriented Programming, DBMS, Operating Systems, distributed systems, scalability, debugging, and SDLC.

EDUCATION

SRM Institute of Science and Technology, Tiruchirappalli - B.Tech, Computer Science and Engineering | Graduating May 2027
| CGPA: 8.23/10

EXPERIENCE

AI Engineering Intern - BitApps | Apr 2026 - Present

- Improved web application release reliability by identifying and resolving UI regressions through automated Playwright testing with Python during CI/CD workflows.
- Reduced manual workflow setup and execution effort by developing a Python/FastAPI backend to programmatically trigger and manage workflow automations.
- Improved development and testing consistency by containerizing automation services with Docker across engineering environments.
- Increased development workflow efficiency by using Git/GitHub feature branches, pull requests, and collaborative version-control practices.
- Contributed to backend development, debugging, automated testing, and SDLC activities to improve overall application quality and maintainability.

PROJECTS

ResolveHub - IT Service Management Platform | Python, FastAPI, PostgreSQL, React, Docker

- Built a full-stack IT service management platform for incidents, service requests, and knowledge articles using FastAPI, PostgreSQL, React, and Docker.
- Secured the platform for 3 user roles - Admin, Agent, and Requester - by implementing JWT authentication and role-based access control.
- Improved persistent data management by developing REST APIs with FastAPI and PostgreSQL.
- Automated incoming ticket categorization by integrating the Google Gemini API, reducing manual triage effort.
- Improved deployment consistency by containerizing application services with Docker and deploying the application on Render.
- Improved application reliability by developing automated backend and frontend test suites to validate core functionality.

Online Auction System | Java, Spring Boot, MongoDB, React, Node.js, WebSockets

- Built a real-time online auction platform supporting concurrent bidding and live bid updates using Spring Boot, MongoDB, React, Node.js, and WebSockets.
- Enabled real-time bid synchronization across connected clients by implementing WebSockets for live bid broadcasting.
- Secured user access and API endpoints by implementing JWT-based authentication.
- Prevented race conditions and maintained bid consistency during simultaneous transactions by implementing MongoDB transactions and concurrency controls.
- Supported concurrent user interactions by designing backend services around real-time communication and bidding workflows.

Intelligent Traffic Management System | React, TypeScript, Machine Learning

- Built an adaptive traffic management system simulating signal control across multiple intersections using React, TypeScript, and machine learning.
- Improved signal responsiveness by applying a machine learning model to forecast congestion and dynamically adjust traffic-signal timing.
- Reduced simulated average waiting time compared with fixed-cycle signal timing by implementing adaptive signal-control logic.

LLM Fine-Tuning Pipeline | Python, PyTorch, Hugging Face, Docker, MLflow

- Built a modular LLM fine-tuning pipeline supporting LoRA-based training across multiple open-source base models.
- Improved experiment reproducibility and model comparison by integrating MLflow and Weights & Biases for experiment tracking.
- Standardized training execution across environments by containerizing the fine-tuning workflow with Docker.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Data Structures & Algorithms: Arrays, Strings, Hashing, Stacks, Queues, Linked Lists, Trees, Graphs, Sorting, Searching, Recursion, Dynamic Programming

Backend & APIs: FastAPI, Spring Boot, Node.js, REST APIs, WebSockets, JWT Authentication

Frontend: React.js, TypeScript, HTML/CSS, Tailwind CSS

Databases: PostgreSQL, MongoDB, MySQL, Redis

Cloud & DevOps: AWS, Docker, CI/CD, Git, GitHub, Postman

Testing & Engineering: Playwright, Pytest, Automated Testing, API Testing, Debugging, Code Reviews

AI/ML: PyTorch, Hugging Face Transformers, LangChain, Scikit-learn, LLMs, Generative AI

Core CS: Object-Oriented Programming, Data Structures & Algorithms, DBMS, Operating Systems, Distributed Systems, Scalability, SDLC

CERTIFICATIONS

- Getting Started with Artificial Intelligence - IBM SkillsBuild, 2026
- MongoDB Developer Path - 12 Micro-Certifications, 2024